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SUBJECT CODE NO: RNP-8009
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (Mechanical) (NEP) Sem - III
Examination May/June-2026
Machine Drawing

[Time: 1:30 Hours] **[Max. Marks: 30]**

Please check whether you have got the right question paper.

N. B
1. Part A: Question No 1 is compulsory.
2. Attempt any four questions from remaining Part B: Q.2 to Q.7.

PART – A

Q.1 Solve any 5 following questions **10**

1) The process of unfolding or unrolling a 3D object into a 2D plane is called:
   A) Sectioning B) Projection
   C) Development D) Isometric drawing

2) The development of a cylinder is in the form of a:
   A) Rectangle B) Triangle
   C) Circle D) Rectangle with two circles as bases

3) An auxiliary view helps to obtain:
   A) Hidden view B) True shape and size
   C) Sectional view D) Simplified drawing

4) A helix is a curve generated by:
   A) A point moving around and along a cylinder simultaneously
   B) A rolling circle on a straight line
   C) A circle on a cone
   D) A parabola

5) A rivet joint is a type of:
   A) Temporary joint B) Permanent joint
   C) Sliding joint D) Flexible joint

6) The key in a shaft is used to:
   A) Prevent relative motion between shaft and hub
   B) Increase torque
   C) Reduce friction
   D) Support load

7) The thread angle of ISO metric thread is:
   A) 30° B) 45° C) 55° D) 60°

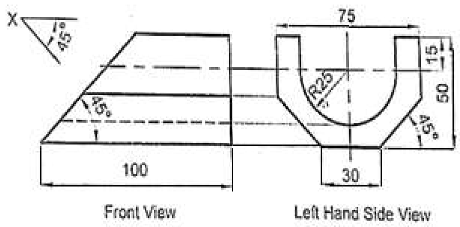
PART – B

Solve any four from the following

Q.2 A hexagonal prism, edge of base 20 mm and axis 50 mm long, rests with its base on H.P such that one of its rectangular faces is parallel to V.P. It is cut by a plane perpendicular to V.P, inclined at 45° to H.P and passing through the right corner of the top face of the prism. Draw the sectional top view and develop the lateral surface of the truncated prism. **5**

Q.3 A cylinder 50mm dia. and 70mm axis is completely penetrated by another of 40 mm dia. and 70 mm axis horizontally Both axes intersect & bisect each other. Draw projections showing curves of intersections. **5**

Q.4 fig shows a front view and left hand side view of an object. Draw the given views and project an auxiliary top view looking in the direction of X. **5**



Q.5 Draw an ellipse of major axis 100mm and minor axis 50mm by concentric circle method. Also draw tangent and normal at any point on the curve. **5**

Q.6 Draw the sectional front view and top view of a knuckle joint. **5**

Q.7 Represent the conventional symbols for steel, brass, and aluminium materials and indicate surface finish symbols. **5**